Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions to guide strategic decisions

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Dataset at a Glance

3,900

Transactions

Rows in dataset

18

Columns

Features tracked

37

Missing Values

Review Rating only

50

Locations

Unique regions

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Data Preparation in Python

Clean

Imputed missing Review Ratings using category medians

Engineer

Created `age_group` bins & `purchase_frequency_days`

Load

Connected to PostgreSQL for SQL analysis

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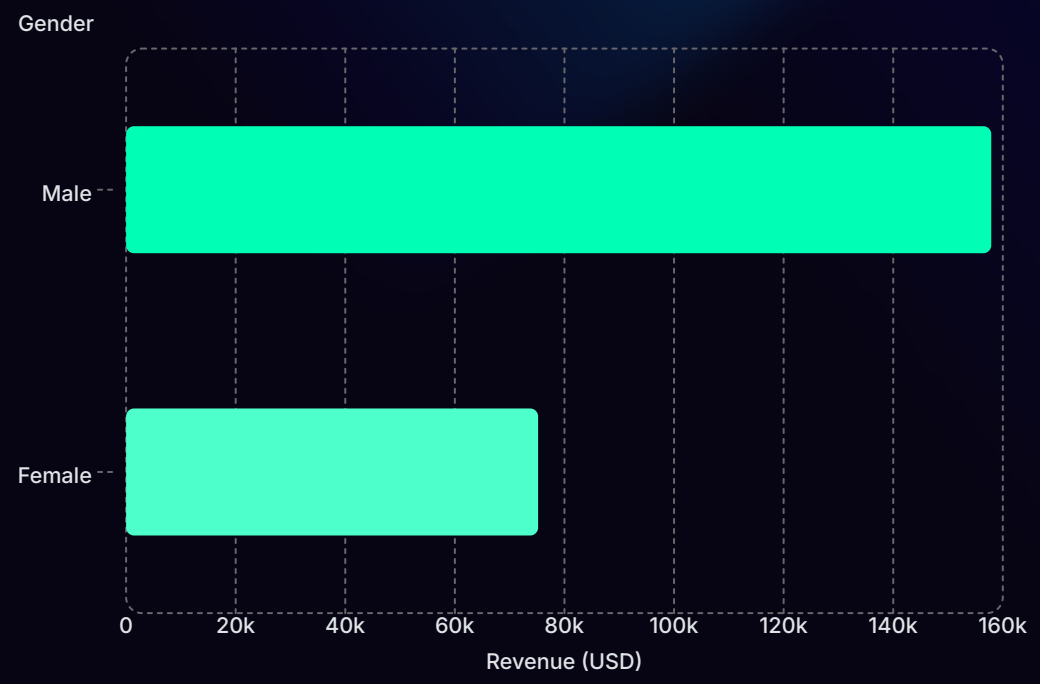
Revenue by Gender



Key Finding

Male customers generate **2.1x more revenue** than female customers

Male: **$157,890**

Female: **$75,191**

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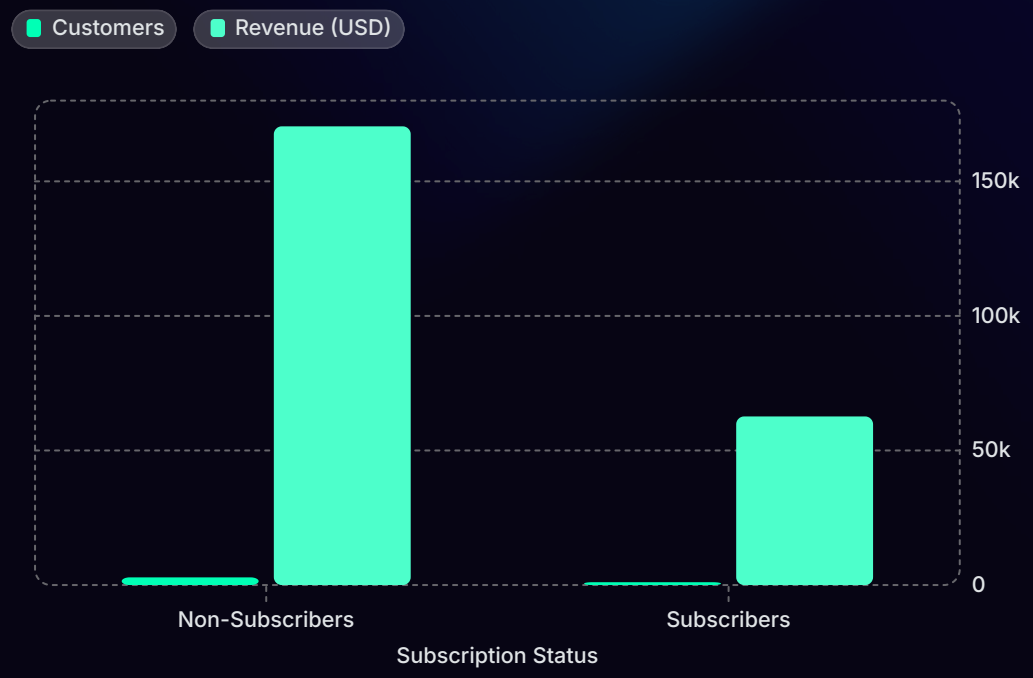
Subscribers vs. Non-Subscribers



Subscription Gap

Only **27%** of customers subscribe

Average spend is nearly equal — opportunity to convert

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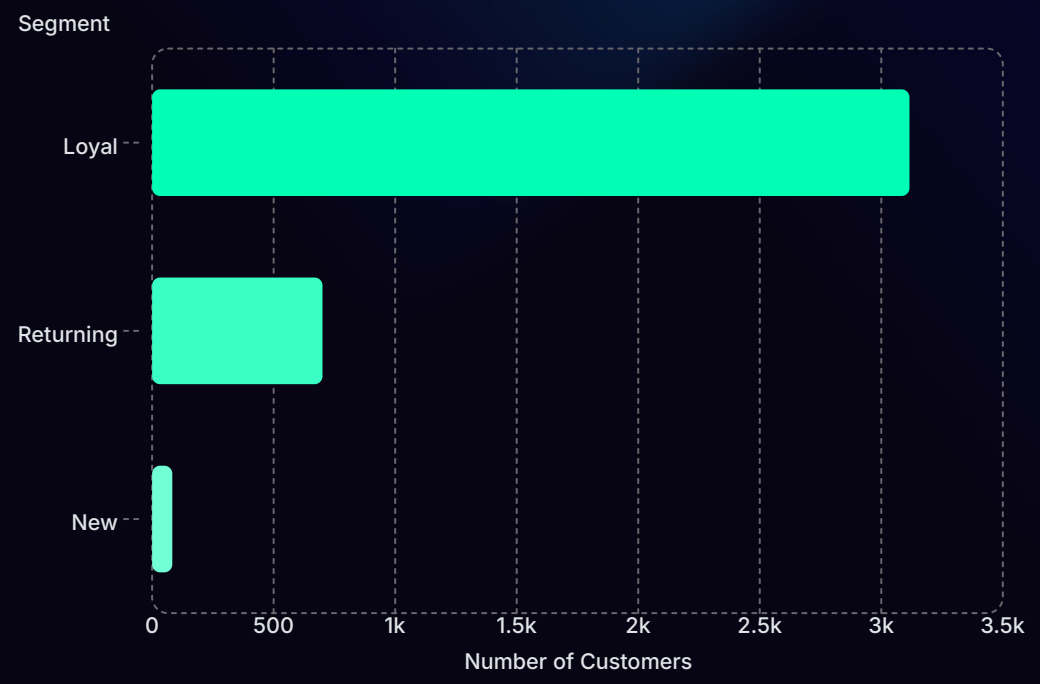
Customer Segmentation



Loyalty Concentration

80% of customers are Loyal

Only 83 New customers — growth opportunity

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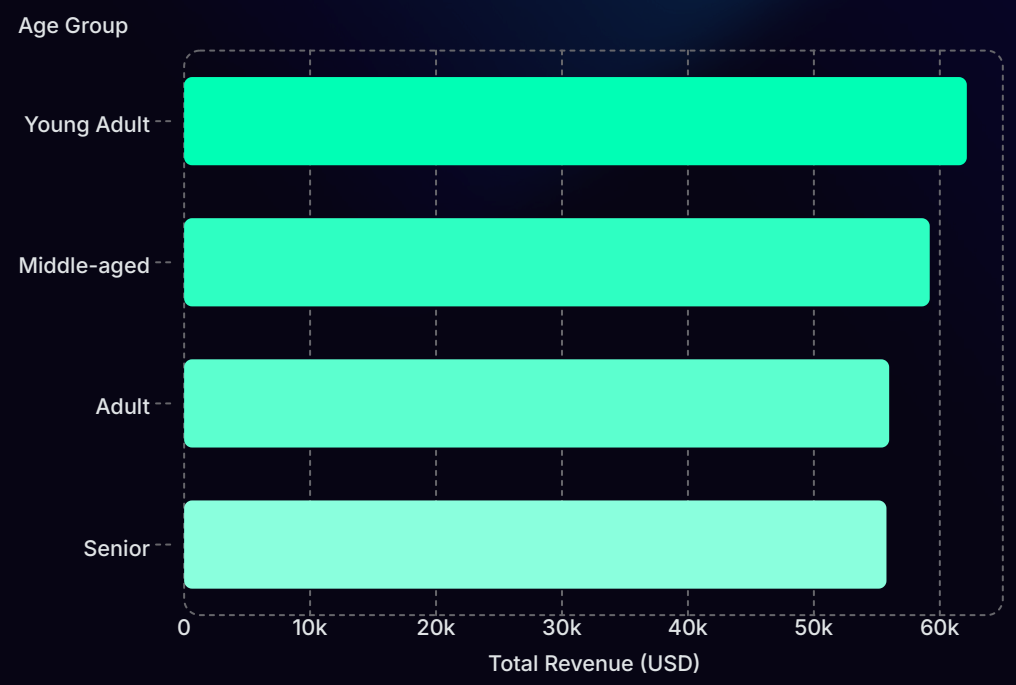
Revenue by Age Group



Age Group Insights

Young Adults lead with **$62,143** in revenue

Revenue is relatively evenly distributed across all age groups

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Top Products & Discount Behavior

Top 5 by Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Product	Discount Rate
Hat	50%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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Power BI Dashboard

3.9K

Total Customers

$59.76

Avg Purchase Amount

3.75

Avg Review Rating

Visualizations include revenue by category, sales by age group, and subscription status breakdown

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discounts

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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